A Geometric Criterion for Sway and Rocking in Inclined Pile Groups

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Abstract

Operational modal analysis of a pile-supported foundation returns natural frequencies and mode shapes, but it does not identify the mechanism that each mode represents. For an inclined pile group the distinction matters most at the lowest frequency, where the cap may sway on the lateral stiffness of the piles or rock on their axial stiffness. These are the two fundamental motions of such a system, and the measured records alone seldom establish which of them has been identified. The conventional remedy is to construct a numerical model of the foundation and to classify the measured modes by correspondence with it. The difficulty is that such a model requires the stiffness of the supporting soil, which is the least certain quantity in the problem: it depends on the strain level, on the installation process and on time. A classification obtained in this way therefore inherits the uncertainty of its least reliable input. This paper establishes the classification analytically instead. Both motions of an inclined pile group are resisted by the same axial pile stiffness, and that stiffness cancels from the ratio of the two natural frequencies. The ratio reduces to a function of the geometry alone, which is available from the setting-out drawing. The criterion is demonstrated on the foundation of an offshore wind turbine, measured before and after tower installation. The single band that satisfies the requirements placed on a physical mode is shown to be sway, in agreement with the geometric prediction.

Keywords: inclined pile group; pile-group foundation; operational modal analysis; sway and rocking; offshore wind turbine; soil–structure interaction

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1 Introduction

A modal identification algorithm fits a linear state-space model to measured covariances and returns its eigenvalues. Nothing in that operation distinguishes a mode of the structure from mains interference or from a resonance of the instrument mounting. All three appear as poles, all three can be made to stabilise cleanly across model orders, and all three can carry excellent quality indices. Stabilisation measures the internal consistency of a fit; it provides no evidence about what was fitted.

Acceptance must therefore be imported from outside the fit. Two sources are available before any measurement is taken. The first is the definition of a mode: a pair of eigenvalue and mode shape satisfying the homogeneous equation of motion, in which the mass, damping and stiffness belong to the structure and not to the instrumentation. The second is the structure of the estimator itself, which can only resolve what its covariance window spans. Both yield requirements that can be applied to raw records without a validated model, a documented sensor layout or environmental data, which matters in construction-stage measurement, where none of the three is normally available.

A parallel difficulty arises on the modelling side. The lowest mode of a pile-group foundation may be a sway of the cap on the lateral stiffness of the piles, or a rocking of the cap on their axial stiffness. The question is usually settled by constructing a model, and such a model requires the soil stiffness. That is the least certain quantity in the problem: it depends on strain level, on installation effects and on time, and for a foundation measured during construction it is rarely known to better than a factor of two. It is shown below that for an inclined pile group the classification does not require it. Both mechanisms are resisted by the same axial pile stiffness, so it cancels from the ratio of their natural frequencies, and that ratio is fixed by geometry and mass distribution alone.

This paper derives a geometric criterion that orders sway and rocking in an inclined pile group, so that a measured band can be assigned to a mechanism without a soil model. It also sets out acceptance requirements that follow from the definition of a mode and can be evaluated on raw records. The two are brought together on a construction-stage measurement of an inclined pile group, where the mechanism predicted from the setting-out drawing proves to be the mechanism the records display.

Section 2 sets both parts against the work on which they draw. Section 3 describes the foundation, the ground and the records. Section 4 derives the criterion from geometry alone.

Section 5 identifies the modes and decides which poles belong to the structure. Section 6 combines the two independent lines of evidence.

2 Related work

Four bodies of work bear on the problem: the stiffness of inclined pile groups, field identification of wind turbine foundations, the acceptance of identified poles, and the contamination of ambient records by harmonic and local responses.

2.1 Stiffness of inclined pile groups

The stiffness model of Section 4 is classical. Hrennikoff [1] analysed a rigid cap on batter piles by resolving the axial and transverse resistance of each pile onto the displacements of the cap, which is the route by which inclination turns axial stiffness into lateral stiffness. Saul [2] cast the analysis as a matrix of stiffness coefficients from which the natural frequencies of a pile foundation also follow, and Poulos and Davis [3] give the standard treatment of pile groups. Randolph and Wroth [4, 5] provide the closed-form axial stiffness of a single pile and the axial interaction factors used in Section 4.2. The benefit of inclination depends on the axial capacity it mobilises: in centrifuge tests McVay et al. [6] found battered groups to carry 20 to 50 % more lateral load than plumb groups in dense sand, but little more in loose sand, where the tension capacity of the piles was limited.

Dynamic interaction within pile groups was established by Kaynia and Kausel [7], who found group impedances to depend strongly on frequency, and reduced to practice through the interaction factors of Dobry and Gazetas [8] and of Gazetas and Makris [9, 10]. For inclined piles, Gerolymos et al. [11] showed that batter piles may be beneficial or detrimental according to superstructure slenderness and pile–cap connection, and Giannakou et al. [12] that the response depends on the balance between overturning moment and shear. Padrón et al. [13] computed the complete vertical, horizontal, rocking and cross-coupled impedance functions of inclined piles and pile groups with a coupled boundary-element and finite-element formulation. The same line of work has since examined the effect of the angle of inclination on the dynamic properties of piled structures [14], extended the impedances to layered soils [15], and provided ready-to-use vertical and rocking impedances of pile groups for offshore wind turbine foundations [16], while Goit and Saitoh [17] measured the interaction factors and impedances of inclined piles experimentally. Huang and Zhong [18] derived the coupled horizontal–rocking impedance of partially embedded pile groups, the configuration of an elevated cap, and showed its effect on the resonance of offshore wind turbines, for vertical piles. Arany et al. [19, 20] derived closed-form estimates of offshore wind turbine frequencies that include foundation flexibility.

These studies predict an impedance, a response or a frequency, and each requires the stiffness of the soil to do so. To the authors' knowledge none forms the ratio of the sway and rocking frequencies of an inclined group, from which the common axial pile stiffness cancels, or uses that ratio to assign a measured mode to a mechanism. The result of Section 4 is narrower than an impedance analysis and less demanding: it gives a ratio in place of an impedance and a classification in place of a response, without the soil model those studies require.

2.2 Field identification of wind turbine foundations

Output-only identification of offshore wind turbines is established. Devriendt et al. [21] automated operational modal analysis to track the support-structure modes of a turbine in the Belgian North Sea, Damgaard et al. [22] identified cross-wind modal properties of offshore turbines by full-scale testing, and Dong et al. [23] combined vibration monitoring with operational modal analysis of an offshore turbine structure. Weijtjens et al. [24] used resonance frequencies to monitor a monopile foundation, and Norén-Cosgriff and Kaynia [25] found the first natural

frequency to fall and damping to rise with load level. Tcherniak et al. [26] examined where the assumptions of the method break down on operating turbines, and van Vondelen et al. [27] reviewed damping identification for offshore turbines. Separating a structural change from environmental and operational variability requires long records with environmental covariates [28, 29], which construction-stage campaigns rarely provide.

Measured frequencies of monopile-supported turbines commonly exceed design predictions. Sastre Jurado et al. [30] found the parked eigenfrequencies of a North Sea wind farm to be systematically underestimated by the design models and identified foundation stiffness as a key source of the discrepancy, and Winkler et al. [31] benchmarked soil–structure interaction models against monopiles measured before the tower was installed. Over 47 locations, Nernheim et al. [32] found design frequencies between about 85 and 97 % of the measured values, a gap that stiffening the soil springs alone did not close. The inclined group studied here shows the opposite sign: the axial pile stiffness inverted from the measurement lies below the forward estimate, and the two are reconciled only when pile–soil–pile interaction is included (Section 6.2).

Measurements during construction are fewer. Fernandes et al. [33] tracked resonance frequencies through every installation stage of 48 monopile-supported turbines, Seo et al. [34] tested a turbine on tripod suction buckets at four construction stages, and Louw and Kearsley [35] monitored the strains of an onshore piled raft through turbine installation and its first year of operation. Pile groups under an elevated cap, the foundation of the Donghai Bridge wind farm in China and of the nearshore farms of the Mekong Delta [36, 37], have been studied mainly for bearing capacity [38]. Li et al. [39] identified the modes of an operating 4 MW turbine on such a foundation, eight piles inclined at 6:1 under a 15 m cap, from accelerometers distributed along the tower, and found the first mode at 0.36 Hz; Liu et al. [40] studied a cap–pile group of the Donghai Bridge farm by experimental modal analysis. In Vietnam, Huynh [41] modelled a single pile foundation as a rigid body to study the loads from blade breakage. These studies identify the modes of the turbine as a whole, measured along the tower or at a single level. None that was found instruments the cap of an inclined group so that its pattern of motion, and not its frequency alone, can be compared with a mechanism predicted beforehand.

2.3 Acceptance of identified poles

Covariance-driven subspace identification [42, 43] and frequency domain decomposition [44] are the established output-only methods, and Reynders [45] reviews them together with the assumptions behind them. Stabilisation is the usual basis for accepting a pole, and the literature that defines it is explicit about what it tests. Peeters and De Roeck [46] describe it as discarding the spurious poles of an over-specified model by requiring agreement with the model one order lower, Döhler and Mevel [47] as separating physical from spurious modes by their constancy across orders, and Zhang et al. [48] judged its power to discriminate limited. Automated procedures add hard and soft validation criteria and cluster the stable poles [49–52]; modal phase collinearity [53] and the modal assurance criterion [54] test the consistency of each shape, and uncertainty bounds quantify the scatter of each estimate [47, 55]. Apart from limits on damping, these tests concern the consistency of the fit and the quality of each pole within one record. They do not test properties that a structural mode must have independently of the fit, such as extending over the instrumented structure or persisting when the sensors are moved.

Invariance across setups enters multi-setup identification as an assumption. Reference-based identification [43] rests on the invariance of an eigenvalue to the sensor configuration, and the merging schemes of Mevel et al. [56] and Döhler and Mevel [57] fit one state matrix to all setups because the same structure is observed in each. Döhler et al. [58] noted that matching modes between setups after identification is fragile and that the assumption may fail in practice. Section 5.1 uses the same invariance as a test that a candidate pole must pass, not as a constraint imposed on the fit.

The number of block rows, which sets the covariance window of the estimator, is known to

bias the estimates [59], and Priori et al. [60] proposed rules for its minimum value. Section 5.1 states the corresponding limit as a bound on the frequencies the estimator can resolve, and Table 6 shows that the bound alone suppresses the admitted band. Subspace identification has been shown to be robust to white excitation whose covariance varies in time [61], but not to impulsive or coloured excitation, under which damping estimates show larger variance and bias [62]; damping is therefore not interpreted here.

2.4 Harmonic and local contamination

A deterministic harmonic can be identified as a structural pole [63], and several indicators separate the two. Brincker et al. [64] proposed an indicator based on the probability density of the response, which Jacobsen et al. [65] and Andersen et al. [66] applied as the kurtosis of band-pass filtered signals, about 1.5 for a sinusoid against 3 for Gaussian noise. Jacobsen [67] noted that a harmonic behaves as a very lightly damped mode, so that damping is itself a harmonic indicator and a valid range of damping in a stabilisation diagram removes harmonics together with non-physical poles. Modified identification algorithms [68], time–frequency decomposition [69] and projection of the periodic part out of the data [70] have also been used, and Motte et al. [71] reviewed the subject. These studies concern the harmonics of rotating machinery and rotors. The harmonics in the present records are electrical, introduced into the measuring chain at the mains frequency through ground loops [72, 73], but they present to the estimator in the same way. The damping requirement of Section 5.1 follows Jacobsen [67], and its bound is set there against tabulated structural damping.

A resonance need not belong to the structure the sensors were placed to characterise. The mounting of an accelerometer alters the response it records [73,74], and local modes at frequencies close to global ones complicate identification, as Abdelnour and Zabel [75] showed for a space truss, separating the two by a modal strain energy indicator. The effective channel count of Section 5.1 is the participation ratio introduced by Bell and Dean [76] to measure the spatial extent of vibrational modes, and localisation of vibration by structural irregularity was established by Hodges [77] and by Pierre and Dowell [78]. No use of the participation ratio as an acceptance criterion in operational modal analysis was found.

Each of the three requirements of Section 5.1 therefore has an antecedent: the damping bound in the harmonic indicators of Jacobsen, invariance across setups in multi-setup identification, and the effective channel count in the participation ratio of localisation physics. What was not found is their joint use as acceptance tests on construction-stage records, or a mechanism criterion derived independently of the records against which the accepted mode can be checked.

3 The offshore wind turbine foundation and the field measurements

The geometry and the soil profile are read from the design drawings and from the borehole sunk at this turbine. The measurement inventory and the excitation characteristics come from the raw records of two ambient vibration campaigns, one recorded before the tower was erected and one after.

3.1 Foundation geometry and ground conditions

The foundation carries turbine T18 of the Dong Hai 1 wind farm, Tra Vinh, Vietnam. It consists of a circular reinforced concrete cap, 20.5 m in diameter and 4 m thick, supported on 34 prestressed high-strength concrete piles of 1.0 m outside diameter and 46 m length, arranged in three concentric rings. The cap soffit is at elevation +5.435 and the seabed at −6.900, so each

pile stands free in water over 12.34 m before entering the ground; the toe is at approximately −40.5.

Two quantities govern the whole of Section 4: the radius of each ring and the inclination of the piles in it. They resolve a force acting along a pile axis onto the horizontal and rotational degrees of freedom of the cap. Nothing else about the pile enters that resolution: not its section, not its material, not the soil around it. The result of Section 4.2 therefore does not depend on the pile section or the concrete grade at all.

The arrangement was read from the pile setting-out sheet and is reproduced in Figure 1. It was then checked against the dimension string printed along the top of the same sheet. Measured inward from the cap edge, its successive offsets are 1250, $543 + 1563 + 894$, $804 + 696 + 1500$ and $1437 + 1563$ mm, which add up to the three ring radii of 9.0, 6.0 and 3.0 m and to the cap centre. The intermediate grid lines decode in the same way, matching $9.0 \cos 20°$, $9.0 \cos 40°$, $6.0 \cos 30°$ and so on. The layout in Table 1 is therefore a decoding of the dimensions written on the drawing rather than a measurement from a scanned image.

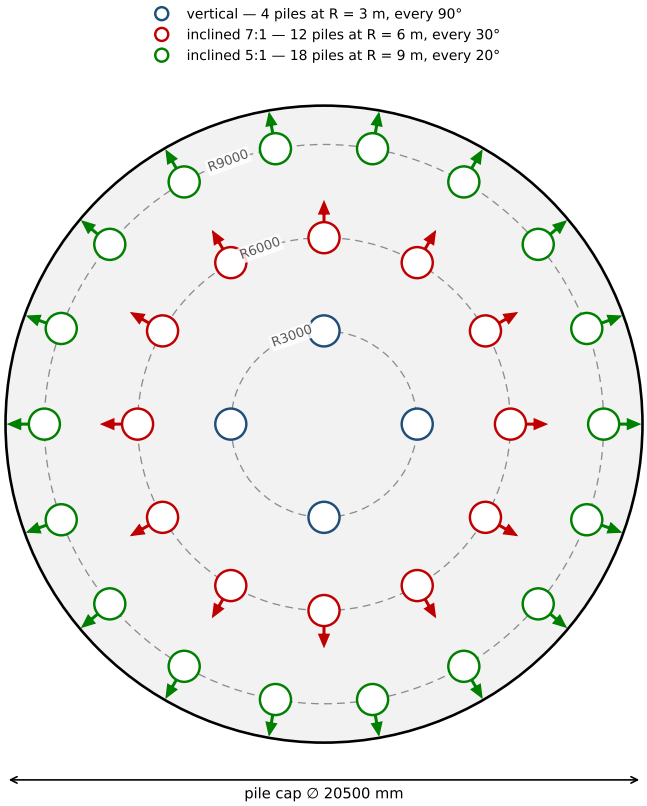


Figure 1: Pile layout of the T18 foundation, redrawn from the setting-out sheet *D1000 PHC pile coordinate*: 34 PHC D1000 piles on three concentric rings of radius 3.0, 6.0 and 9.0 m, carrying 4, 12 and 18 piles respectively. Arrows give the direction of inclination; the inner ring is vertical, the middle ring is inclined 7:1 and the outer ring 5:1. The cap is 20.5 m in diameter

Two lengths determine the dynamic behaviour of the ground beneath the foundation. The pile is unsupported over 12.34 m of water, and the first 13.4 m below the seabed is very soft fat clay with SPT N never exceeding 2, in which it receives almost no lateral restraint. Together these place the whole flexible part of each pile in its upper third. The competent strata below — very stiff lean clay from 16.0 m, medium dense clayey sand to 30.5 m and very stiff fat clay beneath that — act only over the lower half.

This accounts for the fundamental frequency of this foundation falling near 1 Hz rather than

Table 1: Foundation geometry and pile section properties, all read from the design drawings. The section properties enter the absolute frequency of Section 4.2 and the stiffness comparison of Section 6.2; they do not enter the criterion itself.

Cap	20.5 m diameter, 4 m thick; soffit +5.435, top +9.435
Inner ring	$R = 3.0$ m, 4 piles at 90°, vertical
Middle ring	$R = 6.0$ m, 12 piles at 30°, inclined 7:1 (8.1° from vertical)
Outer ring	$R = 9.0$ m, 18 piles at 20°, inclined 5:1 (11.3° from vertical)
Pile	prestressed high-strength concrete, 1.0 m outside diameter, 46 m long; toe at ≈ -40.5
Free length in water	12.34 m (cap soffit +5.435 to seabed −6.900)
Wall thickness	130 mm $\rightarrow A = 0.3553$ m^2, $I = 0.03437$ m^4
Elastic modulus	40 GPa $\rightarrow EA = 1.42 \times 10^{10}$ N, $EI = 1.37 \times 10^{9}$ N m^2

in the range expected of a cap bearing on firm ground. The profile and the SPT record are reproduced in Figure 2.

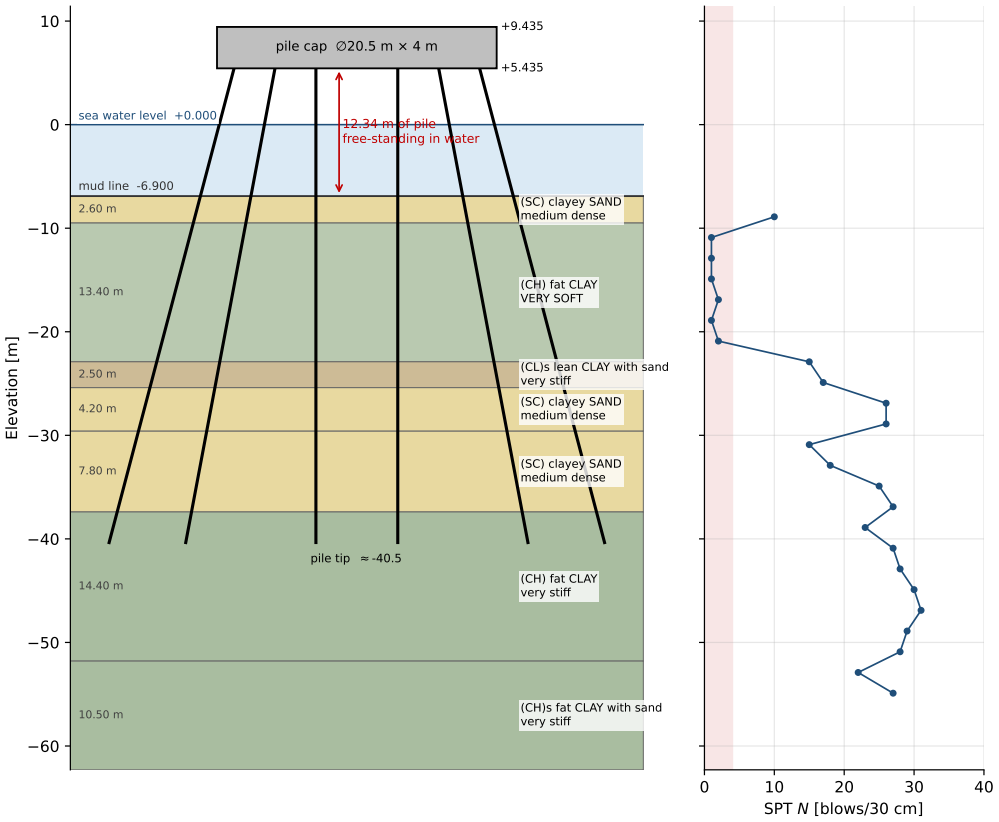


Figure 2: Foundation section and the soil profile at borehole DH9, with the SPT record alongside. Elevations are in metres. Each pile stands free in water over 12.34 m between the cap soffit at +5.435 and the seabed at −6.900. The fat clay immediately below the seabed is very soft, with $N \leq 2$ over 13.4 m

3.2 Instrumentation and measurement campaigns

Two ambient vibration campaigns were recorded on the top surface of the cap: campaign 1 before the tower was erected and campaign 2 after. Each comprises several setups differing in sensor position and measurement direction, seven in total, with three to six accelerometers sampled at 1651 Hz for between 9 and 26 minutes. The instrumented area is a square frame of

5×5 m with sensors at the four corners and at the centre and, in campaign 2, a sixth channel two metres above the centre on the tower itself. Figure 3 shows the arrangement. The position of the frame on the 20.5 m cap is not documented, and it is drawn concentric on the inference that the sixth channel, being mounted on the tower, lies close to the cap centre.

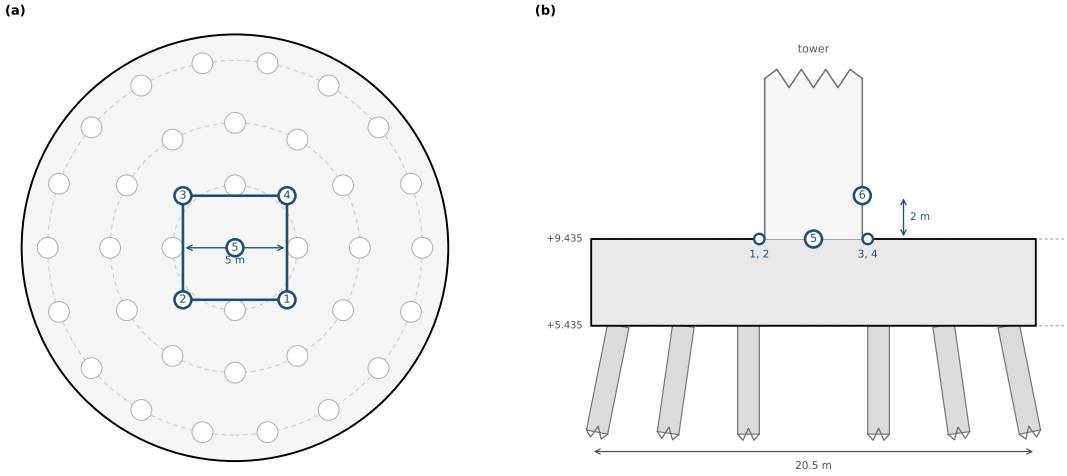


Figure 3: The instrumented frame. (a) Plan of the 5×5 m frame on the 20.5 m cap, with channels 1 to 4 at the corners and channel 5 at the centre. (b) Elevation, showing the sixth channel of campaign 2, which is mounted on the tower two metres above the cap surface and so cannot be placed in the plan. Only the six piles cut by a diametral section are drawn, and both they and the tower are shown broken off rather than at full length. The vertical setups measure Z on every channel; the lateral setups measure horizontally on channels 1, 2, 3 and 5 and vertically on channel 4. The position of the frame on the cap is not documented, and it is drawn concentric on the inference that channel 6, being on the tower, lies close to the cap centre

Table 2: Measurement setups and raw-record statistics. *Peak*/σ is the largest absolute sample of any channel divided by that channel's standard deviation; *RMS drift* the ratio of largest to smallest channel RMS over ten equal segments; *spread* the ratio of the largest to the smallest channel standard deviation within a setup.

Camp.	Setup	Ch.	Samples	Min	Peak/σ	Kurtosis	$> 8\sigma$	RMS drift	Spread
1	Vertical	5	2 535 936	25.6	133	562	2238	3.2	6.1
1	Lateral 1	5	1 523 873	15.4	25	54	3131	12.9	47.0
1	Lateral 2	5	1 385 189	14.0	50	1812	1063	**112**	9.2
1	Three-channel	3	1 545 336	15.6	10	4	9	1.5	33.7
2	Lateral	6	906 399	9.2	**139**	**3883**	724	4.7	11.5
2	Vertical d1	6	1 875 536	18.9	69	95	695	4.4	4.1
2	Vertical d2ke	6	2 260 219	22.8	67	88	**2564**	4.3	4.2

The excitation is not the process the identification method assumes, and the departure is larger in campaign 1 than in campaign 2. Figure 4 shows one campaign 2 record. A single event in it reaches 139 standard deviations of its channel and carries the kurtosis of a nine-minute record to 3883, and the running RMS shows that the event is confined to one channel. Of the signal energy, 78.6 % lies between 50 and 200 Hz and 0.7 % in the structural band from 0.2 to 5 Hz. Table 2 shows that campaign 1 is worse on two counts. In the lateral 2 setup the segment RMS varies by a factor of 112 within one record. In the lateral 1 setup one channel is 47 times weaker than another in the same setup, so that setup carries fewer usable channels than its nominal count suggests. Only the three-channel setup of campaign 1 approaches stationary

broadband behaviour, with a peak of 10σ and a kurtosis of 4.

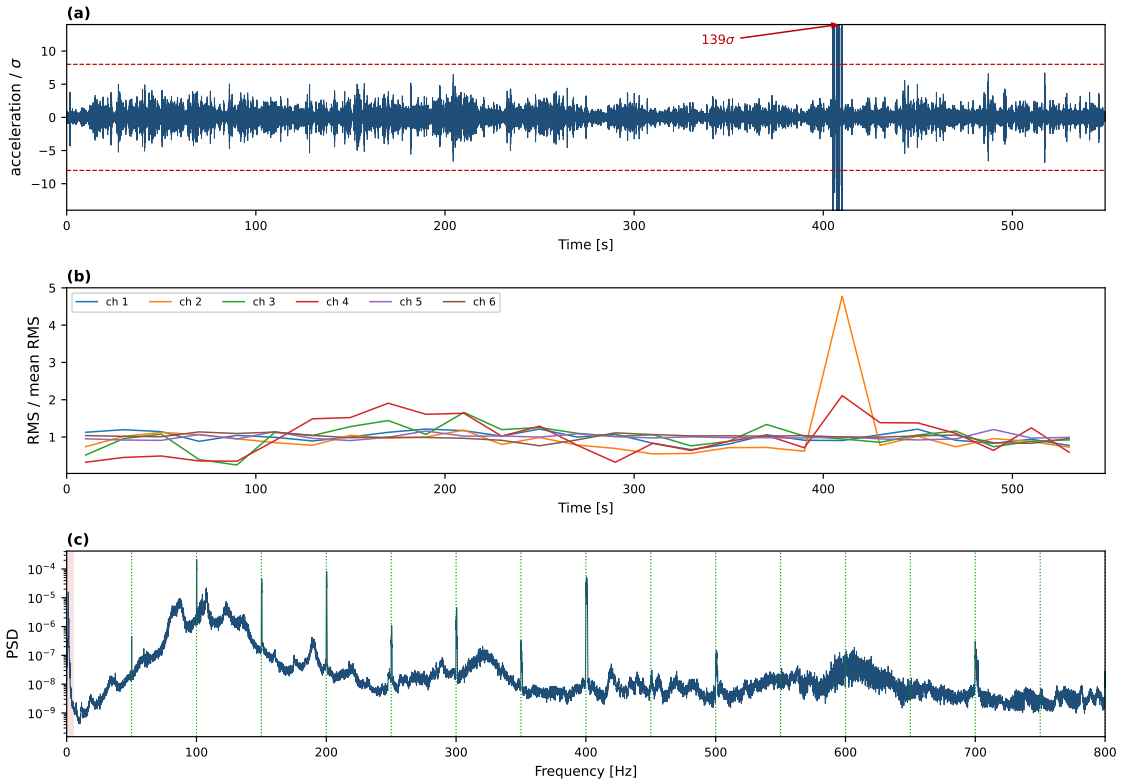


Figure 4: One record of campaign 2 in three views. (a) The full time history of the channel with the largest peak-to-standard-deviation ratio, in units of that channel's standard deviation and with the $\pm 8\sigma$ band marked. The vertical scale is clipped at $\pm 14\sigma$ so that the ambient level remains legible; the single event at 408 s reaches 139σ, carries the record kurtosis to 3883, and is marked by the arrow. (b) The running RMS of every channel over 20 s windows, normalised by its own mean; the event of panel (a) is confined to one channel. (c) The channel-averaged spectrum, with multiples of 50 Hz dotted; the band from 50 to 200 Hz carries 78.6 % of the energy and the shaded structural band from 0.2 to 5 Hz carries 0.7 %

Two consequences follow. Damping ratios are reported where the identification produces them, because the stabilisation criterion uses them, but they are not interpreted. Subspace damping estimates under impulsive, non-stationary excitation are biased by an unknown amount, and field studies of offshore support structures report damping that varies with load level and operating state [25]. Frequency estimates, in addition, carry more uncertainty than the scatter within a stabilisation cluster suggests, and for that reason Section 5.2 does not rest on frequency agreement between setups.

4 A geometric criterion for sway and rocking

The derivation that follows uses no measurement. It starts from the equation of motion of a rigid body supported on axial springs, assembles the lateral and rotational stiffness of an inclined pile group, and shows that the axial pile stiffness cancels from the ratio of the two natural frequencies. Because the result is fixed before any record is examined, the comparison in Section 6.1 is a test and not a fit.

4.1 Idealisation and the stiffness of an inclined pile group

The cap is treated as a rigid body and each pile as a spring acting along its own axis, together with a transverse spring representing its bending. The idealisation is appropriate here because the cap is 4 m thick over a 20.5 m span while the piles are 1.0 m in diameter over 46 m: the cap is orders of magnitude stiffer in the relevant deformation than the elements supporting it. Under this idealisation the foundation has the two degrees of freedom that matter: a translation of the cap, and a rotation about a horizontal axis through its centre. Which mechanism governs the lowest mode is then a question about two stiffnesses.

Resolving the axial stiffness of a single pile onto those degrees of freedom introduces the angle of inclination twice, in two different ways, and this is the origin of the result. Consider a pile at radius r and azimuth θ, inclined outward at an angle α from vertical, so that its axis has unit vector $(\sin\alpha\cos\theta,\ \sin\alpha\sin\theta,\ \cos\alpha)$. Under a horizontal translation u of the cap, the axial extension of that pile is the projection $u\sin\alpha\cos\theta$: only the component of cap displacement lying along the pile axis stretches it, so the contribution to lateral stiffness carries $\sin^2\alpha$. Under a rotation φ of the cap about a horizontal axis, the pile head is displaced vertically by $\varphi\, r\cos\theta$, of which the component along the pile axis is $\varphi\, r\cos\theta\cos\alpha$, so the contribution to rotational stiffness carries $r^2\cos^2\alpha$. Summing over the group and adding the bending term,

$$k_{\mathrm{lat}} = k_{\mathrm{ax}} \sum_i \sin^2\alpha_i + \frac{12EI}{L_{\mathrm{eff}}^3} \sum_i \cos^2\alpha_i, \tag{1}$$

$$k_{\theta} = k_{\mathrm{ax}} \sum_i r_i^2 \cos^2\alpha_i, \tag{2}$$

in which k_{ax} is the axial stiffness of one pile and L_{eff} its effective fixity length.

Figure 5 shows the two mechanisms and the sense in which each loads the piles. Two features of the formulation matter. First, both mechanisms load the piles axially, and in both the axial displacement field is antisymmetric in azimuth, proportional to $\cos\theta$ in each case; this becomes important when pile-group interaction is considered in Section 4.2. Second, the formulation already marks a boundary of the result: a group of vertical piles has $\sin\alpha = 0$ and therefore no axial contribution to lateral stiffness. For such a group, sway is resisted by pile bending alone, the two stiffnesses are no longer proportional to a common quantity, and the criterion derived next does not apply.

4.2 Cancellation of the soil stiffness

Both stiffnesses in (1) and (2) are proportional to the same axial pile stiffness. Setting the bending term aside for the moment, k_{ax} cancels when the ratio of the corresponding natural frequencies is formed:

$$\frac{f_{\mathrm{rock}}}{f_{\mathrm{sway}}} = \sqrt{\frac{k_\theta/J}{k_{\mathrm{lat}}/M}} = \sqrt{\frac{\sum r_i^2\cos^2\alpha_i/J}{\sum \sin^2\alpha_i/M}}, \tag{3}$$

in which J is the mass moment of inertia of the foundation about the rotation axis and M its mass. Nothing on the right-hand side refers to the soil, to the pile section or to the pile material. For a group in which all piles share one angle of inclination the expression simplifies further. Writing $\sum r^2 = n\, r_{\mathrm{rms}}^2$ and $\sum \sin^2\alpha = n\sin^2\alpha$, and $k_g = \sqrt{J/M}$ for the radius of gyration of the mass,

$$\frac{f_{\mathrm{rock}}}{f_{\mathrm{sway}}} = \frac{r_{\mathrm{rms}}}{k_g}\,\cot\alpha, \tag{4}$$

two dimensionless numbers: the reach of the pile ring relative to the inertia of the mass it supports, and the inclination.

This form makes the range of validity explicit. The two mechanisms coincide when $\tan\alpha = r_{\mathrm{rms}}/k_g$. For a cap whose piles reach beyond its radius of gyration, as any practical group

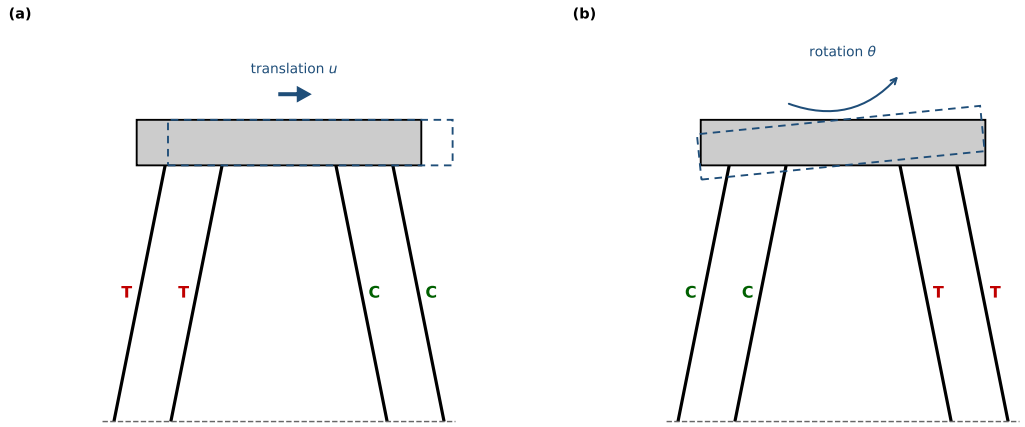


Figure 5: The two mechanisms available to a rigid cap on an inclined pile group. (a) Sway, resisted by the axial stiffness of the inclined piles together with pile bending. (b) Rocking, resisted by the same axial stiffness acting on the lever arms. Dashed outlines give the displaced position; C marks a pile in compression and T a pile in tension

does, that angle lies above 45°; for the foundation studied here it is 52°. Inclinations used in construction are an order of magnitude shallower, between 12:1 and 4:1, or 4.8° to 14.0°, so the ratio does not fall below about five for any practical inclined group, irrespective of the soil. Figure 6 shows the relation over the practical range for several values of r_{rms}/k_g. A single low-frequency band identified on such a foundation can only be sway.

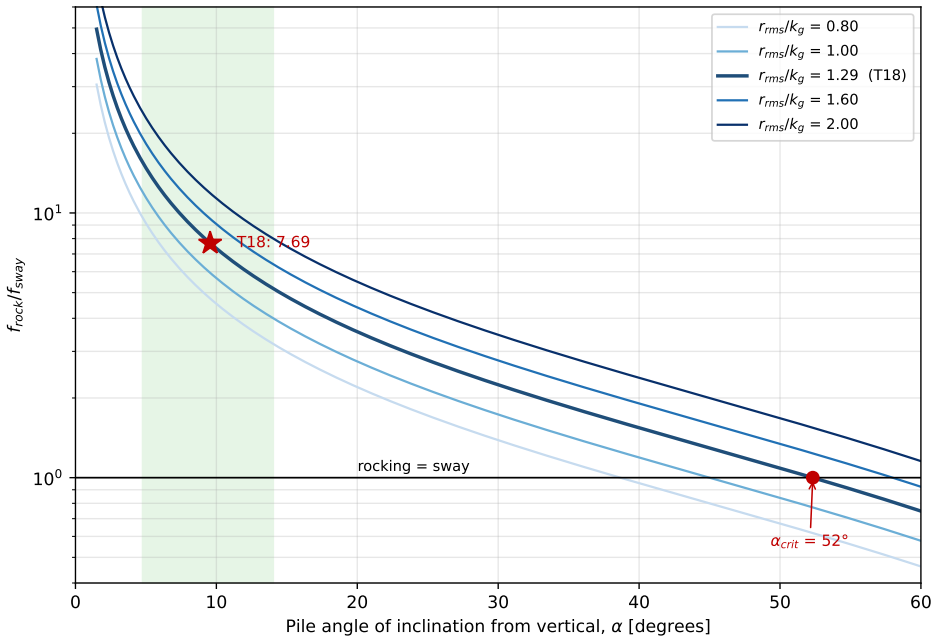


Figure 6: The criterion (4) in dimensionless form, for several values of r_{rms}/k_g. The shaded band covers the inclinations used in construction, 12:1 to 4:1. The horizontal line is $f_{\mathrm{rock}} = f_{\mathrm{sway}}$, reached only at the marked angle α_{crit}, which lies far outside that band

One mechanism could break the cancellation. Pile–soil–pile interaction reduces the stiffness of a group below the sum of its isolated piles, and it would not cancel if the two mechanisms

mobilised the soil differently. It is therefore computed here rather than assumed away. Elastic axial interaction factors [5], $\alpha_{ij} = \ln(r_m/s_{ij})/\ln(r_m/r_0)$ with $r_m = 2.5\,L_{\mathrm{emb}}(1-\nu)$, are assembled over the 34 actual pile positions. For a rigid cap, a mechanism whose unit axial displacement vector is $\mathbf{v}$ then has group efficiency $\eta = (\mathbf{v}^{\mathsf{T}}\mathbf{A}^{-1}\mathbf{v})/(\mathbf{v}^{\mathsf{T}}\mathbf{v})$. The effect very nearly cancels as well, for the reason identified in Section 4.1: both mechanisms produce axial displacement fields proportional to $\cos\theta$, so in both the interactions between piles on opposite sides of the cap oppose one another. The efficiencies are 0.447 for sway and 0.399 for rocking, against 0.075 for uniform settlement, in which all piles move in the same sense and the interactions accumulate instead. The ratio is corrected by $\sqrt{0.399/0.447} = 0.945$.

Table 3 collects the geometric sums and the resulting values for this foundation. The ratio is 7.69 with the group treated as isolated piles and 7.26 with interaction included, and the bending term, when restored, moves it between 6.15 and 7.48 over a tenfold range of assumed axial stiffness. Rocking lies approximately seven times above sway on this foundation, and no plausible variation of the assumptions brings it within a factor of five.

Table 3: Geometric sums and the resulting frequency ratio for the T18 group.

$\sum r^2 \cos^2\alpha$	1861.3 m^2
$\sum \sin^2\alpha$	0.9323
$\sum \cos^2\alpha$	33.07
M (cap, piles and added water)	5.03×10^6 kg
J	1.698×10^8 kg m^2
r_{rms} , k_g	7.53 m , 5.81 m
Group efficiency: sway / rocking / uniform	0.447 / 0.399 / 0.075
$f_{\mathrm{rock}}/f_{\mathrm{sway}}$, isolated piles	**7.69**
$f_{\mathrm{rock}}/f_{\mathrm{sway}}$, with group interaction	**7.26**
Inclination at which the mechanisms coincide	52°

The cancellation is a first-order result under the idealisation of Section 4.1, not a universal property of inclined pile groups, and the limits belong with it. It assumes a rigid cap, small motion, a single effective axial stiffness entering both mechanisms, and static rather than frequency-dependent pile stiffness. Full impedance analyses of inclined pile groups show that group stiffnesses depend on excitation frequency, on the soil–pile stiffness ratio, on spacing and on direction of inclination, and that horizontal–rocking cross-coupling can be significant and can change sign with configuration [13]. The ordering survives those complications; the numerical value does not. The two mechanisms are separated by a factor of seven under the present idealisation, whereas the corrections a full impedance treatment would introduce are of order tens of per cent. The criterion should be read as a screening test for which mechanism to expect, to be set aside where a full impedance model is available.

The section does not give the absolute frequency. That depends on k_{ax}, and therefore on the soil and on the effective fixity length, neither of which is known here to better than a factor of two; sweeping them spans 1.06 to 2.13 Hz, as Figure 7 shows. The model answers which mechanism governs, not at what frequency it appears.

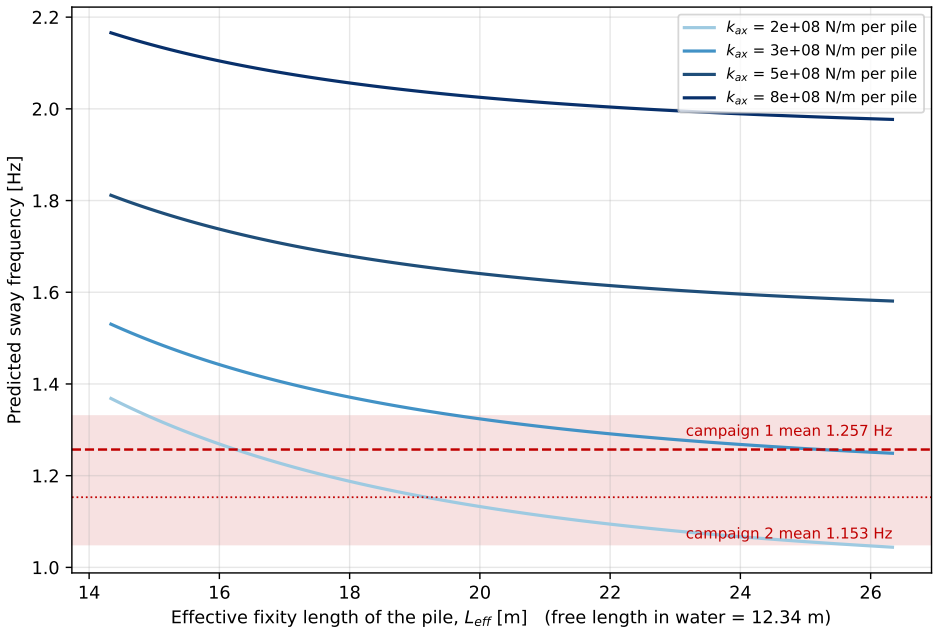


Figure 7: Sensitivity of the predicted sway frequency to the two governing assumptions, the effective fixity length L_{eff} and the axial pile stiffness k_{ax}. The horizontal lines are the campaign means and the shaded strip the measured band, 1.05 to 1.33 Hz

5 Modal identification

Identification supplies candidate poles; it does not supply modes.

5.1 Identification procedure and acceptance criteria

Covariance-driven subspace identification assembles a block Toeplitz matrix of output covariances up to lag $2i$ and recovers the state and output matrices from its singular value decomposition [42, 43]. The estimator therefore observes the record through a time window $\tau = i\,\Delta t$, and this is a property of the construction rather than a processing choice. A mode of period T enters the covariances through their oscillation and decay across that window; when $\tau \ll T$ the covariance varies negligibly over the observed lags and carries nothing that distinguishes the mode from a constant offset. The lowest frequency the estimator can resolve is fixed, for a given configuration, before the data are consulted.

The processing chain follows from that requirement. Retaining the acquisition rate of 1651 Hz with $i = 150$ gives $\tau = 0.091$ s, which is 9 % of one period at 1 Hz and places any such mode outside the span of the estimator. The records are therefore decimated in two stages to 51.59 Hz with zero-phase anti-alias filtering, then high-pass filtered at 0.3 Hz with a fourth-order Butterworth. Identification uses $i = 120$ over model orders 2 to 80, which gives $\tau = 2.33$ s, or 2.3 periods at 1 Hz. Poles are accepted into a cluster when successive orders agree to within 1 % in frequency, 5 % in damping and 0.98 in modal assurance criterion, and a cluster is retained when it contains at least eight poles. A frequency domain decomposition of the same records [44] provides an independent check. Section 5.2 tests the resolution argument directly.

Three requirements are used as acceptance criteria. Each follows from the definition of a mode rather than from a convention of practice, each is stated as a property of the structure, and each can be evaluated on the raw records alone.

The first requirement is that a mode moves more than one measurement point. The mode shape is defined over the structure. If the observed shape has support on one channel and noise elsewhere, then the motion is confined to a region smaller than the sensor spacing, and the pole

describes that region rather than the structure the instrumentation was placed to characterise. The measure used is the effective channel count $N_{\mathrm{eff}} = (\sum |\phi|^2)^2 / \sum |\phi|^4$, taken from the first singular vector of the cross-power spectral density matrix. It equals the number of channels when the shape is spread evenly over them, and unity when one channel carries the entire response. The rejection threshold is established by simulation. Synthetic records were generated at the same length and spectral resolution as the measurements, once with a narrow-band signal on a single channel and once with the same signal shared by all channels in a shape containing a nodal line. The first case gives $N_{\mathrm{eff}} \leq 1.00$ without exception. The second gives ≥ 1.53 for five channels and ≥ 1.74 for six, in 95 % of trials. A threshold of $N_{\mathrm{eff}} = 1.25$ separates them with no overlap in 400 trials per case. The quantity is the inverse participation ratio, familiar from studies of vibrational localisation in disordered media [76, 77]. No precedent was found for its use as an acceptance criterion in operational modal analysis, so it is defined explicitly here and justified by the simulations above rather than by citation. It is not a mass participation factor and should not be interpreted as one.

The second requirement is that a mode is the same for every observer. The eigenvalue depends on the mass, damping and stiffness of the structure, not on which degrees of freedom are instrumented; moving the sensors changes which components of the shape are observed and cannot change the eigenvalue. The criterion requires the same pole in every setup of a campaign and, where the measured degrees of freedom coincide, agreement of the shapes by the modal assurance criterion.

The third requirement is that a mode dissipates at a rate materials permit. Free decay follows an envelope $e^{-\xi\omega t}$, so a mode falls to $1/e$ of its amplitude in $1/(2\pi\xi)$ cycles. At $\xi = 0.01\,\%$ that is about 1600 cycles; at $\xi = 0.007\,\%$ and 100 Hz it is about 2300 cycles, or 23 seconds of sustained oscillation. A built structure under ambient excitation does not sustain free vibration for thousands of cycles, whereas a deterministic harmonic of constant frequency and amplitude presents to a covariance-driven estimator as exactly such a pole, its apparent damping reflecting estimator resolution rather than dissipation [63, 67]. The screening bound used here, $\xi = 0.1\,\%$, is adopted on that basis. Published field damping estimates for offshore support structures vary with load level and operating state [25], so no field value is adopted. The bound coincides instead with the lowest damping that EN 1991-1-4 tabulates for any structural element, a logarithmic decrement of 0.006 for parallel cables, against 0.03 to 0.10 ($\xi \approx 0.5$ to $1.6\,\%$) for reinforced concrete structures [79]. It is used only to reject, and it lies two orders of magnitude below the values the admitted band returns.

5.2 Classification of the identified poles

Applied to these records, the procedure returns poles in three well-separated regions. There is a group between 1.05 and 1.33 Hz and a set between 14 and 24 Hz. A third set, between 57 and 700 Hz, appears only under a configuration that retains the acquisition rate. Table 4 lists the accepted poles of the first two by setup. Figure 8 shows the stabilisation diagram of one setup and illustrates the difficulty: both the 1.2 Hz pole and the 18–20 Hz cluster stabilise cleanly over the full range of model orders, so nothing in the stabilisation behaviour distinguishes them.

Under the first requirement, Figure 9 gives the effective channel count for both bands in all seven setups. In the 13–24 Hz band five of the seven fall at or below 1.17, four of them at exactly 1.00: one channel responds and the rest lie at noise level, so there is no spatial pattern to interpret. In the 1.0–1.4 Hz band six of the seven exceed the threshold, ranging from 1.55 to 5.00. The low-band values are not uniform, and uniformity is not to be expected, since a pattern containing a nodal line necessarily leaves channels near that line with small amplitude, which lowers N_{eff} without implying locality. Two exceptions are reported rather than excluded. Lateral setup 2 of campaign 1 gives 1.00 in the low band as well; it is also the record with the strongest impulsive content, its narrow-band kurtosis [65] lying between 128 and 164. The three-channel setup gives 0.60 of its channel count in both bands, which with only three channels does not

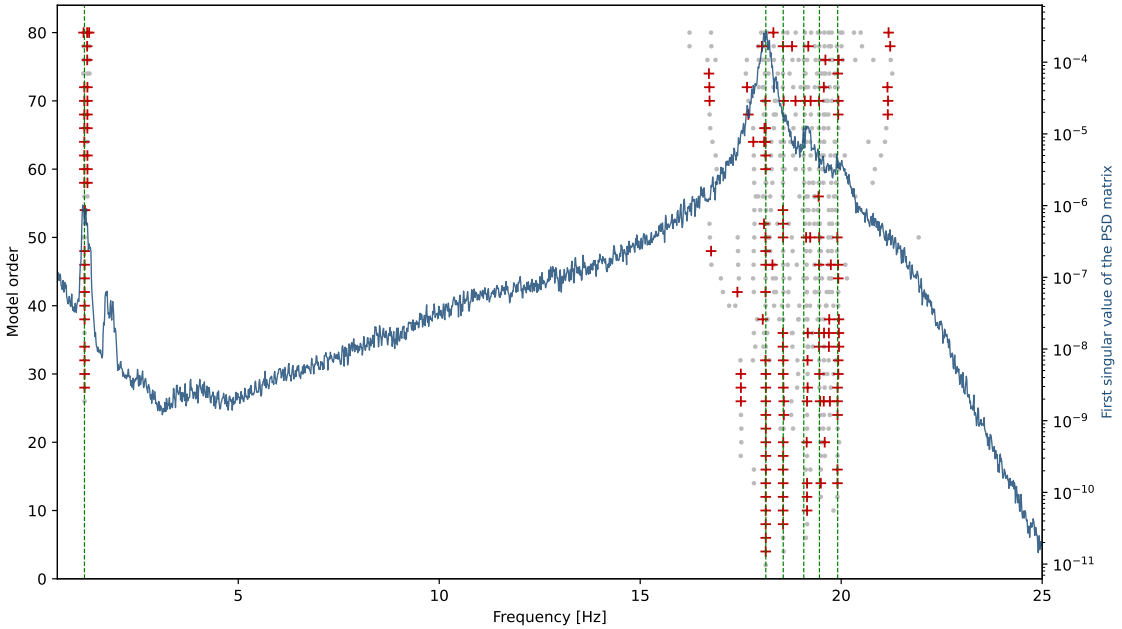


Figure 8: SSI-COV stabilisation diagram for the vertical setup of campaign 1, with the first singular value of the cross-power spectral density matrix overlaid. Both the 1.2 Hz pole and the 18–20 Hz cluster stabilise cleanly over the full range of model orders, so the stabilisation behaviour alone does not distinguish them

Table 4: Accepted poles by setup; frequency in Hz with damping ratio in per cent in parentheses.

Camp.	Setup	Low band	Higher band
1	Vertical	1.173 (4.74)	18.125 (0.70), 18.558 (0.68), 19.070 (0.58), 19.459 (0.59), 19.913 (1.30)
1	Lateral 1	1.226 (3.43), 1.254 (2.77), 1.730 (5.49)	21.912 (0.40), 22.742 (0.51), 23.745 (0.30)
1	Lateral 2	1.322 (2.70)	21.566 (5.95), 22.804 (1.25), 23.164 (0.59)
1	Three-channel	1.237 (3.01), 1.331 (4.39), 1.725 (5.35)	14.432 (1.42), 14.990 (0.97)
2	Lateral	1.052 (2.68), 1.215 (1.93), 1.611 (2.42)	none accepted
2	Vertical d1	1.147 (9.85)	11.801, 13.866, 16.827, 17.081, 17.666, 18.947, 19.466, 19.682, 20.542
2	Vertical d2ke	1.073 (1.25), 1.198 (2.35), 1.230 (2.48)	2.295 (1.19)

separate the two cases.

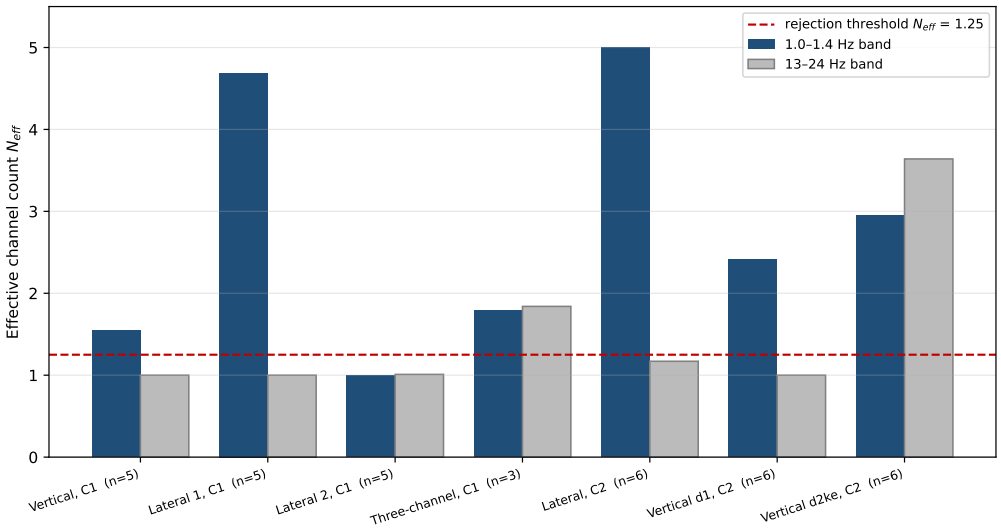


Figure 9: Effective channel count N_{eff} by setup and band, against the rejection threshold of 1.25 established by simulation in Section 5.1. The number of channels in each setup is given in the axis label. A value of unity means that one channel carries the entire response

Under the second requirement, the low-band poles within a campaign spread by ±6.3 % and ±7.7 %. That is an order of magnitude more than the scatter within a stabilisation cluster, so frequency agreement alone cannot settle the question. The criterion is therefore applied to the shapes. In all three vertical setups of both campaigns, opposite corners of the instrumented area move in antiphase, by 176°, 175° and 179°, about a nodal line through the centre of that area. In the lateral setups all horizontal channels move in phase and with similar amplitude. The two vertical setups of campaign 2 instrument five points in common, and between their corresponding clusters the modal assurance criterion is 0.967. The sixth channel of each sits at a different point and is left out of that comparison. Table 5 gives the phase relations and Figure 10 the patterns. One further observation supports this interpretation. The single vertically oriented channel in the lateral setup is also the only channel there that is out of phase and of small amplitude, and it was not selected for that purpose.

By contrast the 14–24 Hz poles fail the same test. Within campaign 1 alone, three setups on the same cap in the same period return three disjoint bands: 18.1 to 19.9 Hz, 21.6 to 23.7 Hz and 14.4 to 15.0 Hz. A natural frequency is a property of the structure and cannot take three values at once.

One observation tells against the interpretation just given. If instead of matching corresponding clusters one compares the first singular vector at the dominant low-band peak of each of those two setups, the modal assurance criterion is 0.09, not 0.97. The peaks are at 1.209 and 1.228 Hz, and the two setups are therefore dominated by different members of the closely spaced group. The shapes of corresponding poles agree; the dominant pole differs between setups. Both facts are consequences of the group containing more than one mode, and Section 6.1 returns to them.

The third requirement needs the influence of the processing configuration settled first. Two processing choices govern what the identification can return, the covariance window and the high-pass cut-off, and varying them together would not separate their effects. One record was therefore processed under all four combinations, Table 6. Configuration C removes the filter while keeping the short window and still returns nothing below 5 Hz; configuration D lengthens the window while keeping the filter and likewise returns nothing. Each constraint is independently sufficient to suppress the admitted band, and only their joint removal recovers it. The resolution

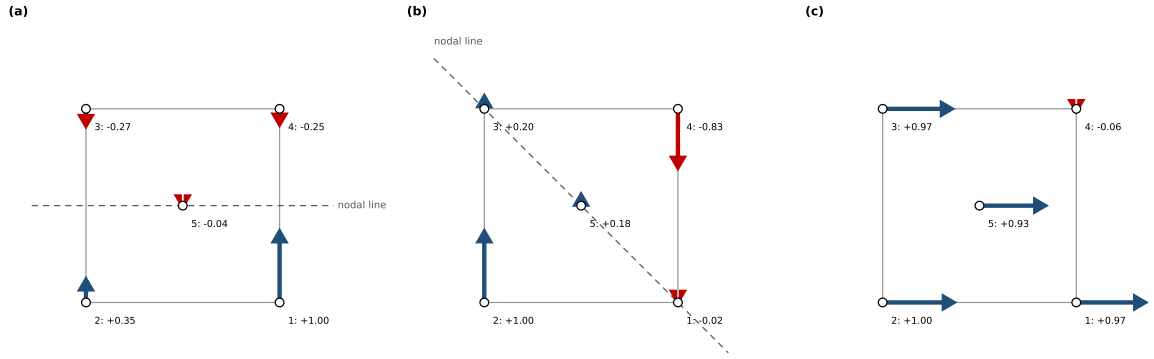


Figure 10: Measured deformation pattern of the 1.05–1.33 Hz band, taken from the first singular vector at the band peak. (a) Vertical setup of campaign 1. (b) Vertical setup d2ke of campaign 2. (c) Lateral setup of campaign 2. Signed amplitudes are given beside each node. The vertical setups show opposite corners in antiphase about a nodal line; the lateral setup shows the horizontal channels in phase, channel 4 being the only vertically oriented channel there

Table 5: Phase between opposite corners of the instrumented frame at the low-band peak, and amplitude at the centre relative to the mean of the corners. In the lateral setup, channel 4 is the only vertically oriented channel, which is why the 2–4 pair behaves unlike the 1–3 pair there.

Setup	Direction	Peak (Hz)	Node 1 vs 3	Node 2 vs 4	Centre / mean corner
Vertical, campaign 1	vertical	1.134	−179.1°	−173.3°	0.048 / 0.466
Vertical d1, campaign 2	vertical	1.209	+176.3°	+176.6°	0.424 / 0.517
Vertical d2ke, campaign 2	vertical	1.228	−142.7°	−175.2°	0.183 / 0.518
Lateral, campaign 2	horizontal	1.203	−4.9°	−143.3°	0.927 / 0.757

limit of Section 5.1 is thereby established as a constraint in its own right and not a by-product of filtering.

Table 6: Factorial comparison of processing configurations, one record.

Config.	Sampling rate	High-pass	i	τ	Poles returned
A	1651 Hz	5 Hz	150	0.091 s	20, from 57.6 to 700.7 Hz; none below 5 Hz
C	1651 Hz	0.3 Hz	150	0.091 s	19, from 80.8 to 604.4 Hz; none below 5 Hz
D	51.6 Hz	5 Hz	120	2.33 s	6, from 6.8 to 20.6 Hz; none below 5 Hz
B	51.6 Hz	0.3 Hz	120	2.33 s	3: 1.052, 1.215, 1.611 Hz

Seven of the twenty poles returned by configuration A carry damping between 0.007 % and 0.042 %, which the third requirement alone rejects. Only afterwards is the spectrum consulted, and it identifies them: the same seven lie within 0.16 % of an exact multiple of 50 Hz, at 100.129, 150.239, 200.209, 250.171, 300.377, 400.537 and 700.666 Hz, and six of them survive into configuration C. They are harmonics of the mains supply, which these records carry at high level, as Figure 11 shows.

The region below the analysis band remains to be examined. The band searched begins at 0.5 Hz, which would exclude a tower mode: soft-stiff design places the fundamental mode of a tower of this class between the $1P$ and $3P$ rotor harmonics, near 0.3 Hz [20]. The records were therefore reanalysed with a further decimation to 12.9 Hz and $i = 150$, a window of 11.6 s or 3.5 periods at 0.3 Hz, and searched from 0.18 Hz. No pole below 0.5 Hz satisfies the criteria in any

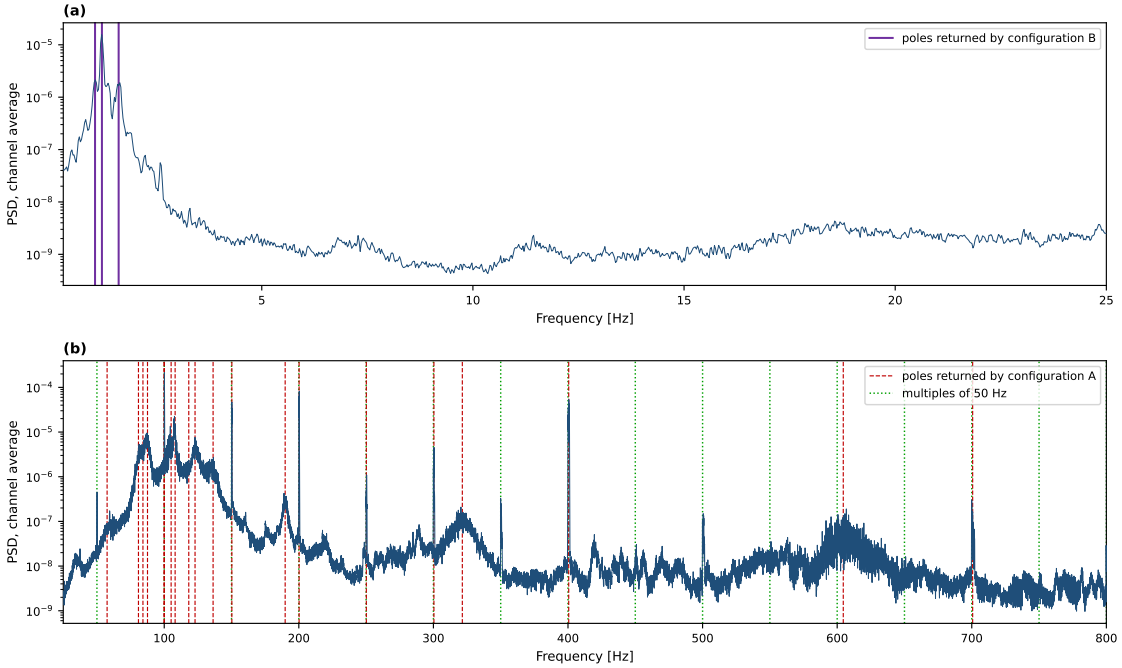


Figure 11: The same record seen by the two processing configurations of Table 6, over (a) the structural band from 0.3 to 25 Hz and (b) the band from 25 to 800 Hz from which configuration A draws its poles. Dotted lines mark multiples of 50 Hz

setup of either campaign. The only features there are peaks near 0.21 Hz in campaign 1 and 0.32 Hz in campaign 2, each consistent between the setups of its own campaign. Their periods are 4.7 s and 3.2 s, and they move upward between campaigns, which no added-mass mechanism can produce. They are attributed to wave excitation under differing sea states. Figure 12 shows the search.

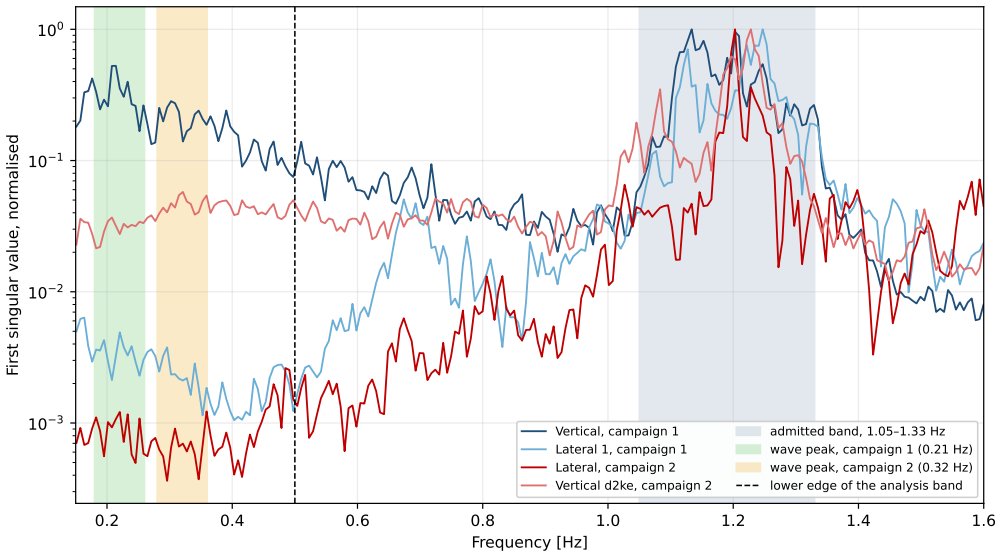


Figure 12: Search below the analysis band, with the records decimated to 12.9 Hz and a covariance window of 11.6 s. The only features below 0.5 Hz are the peaks marked at 0.21 Hz in campaign 1 and 0.32 Hz in campaign 2, which move upward between campaigns and are attributed to wave excitation

Table 7 summarises the three requirements applied together. One band survives all three requirements; the other two fail on grounds independent of each other and of the identification settings.

Table 7: Classification of the three groups of poles.

Group	N_{eff}/n	Same in every setup	Damping	Verdict
1.05–1.33 Hz	0.31–0.94 in six of seven	yes; MAC 0.967 between corresponding clusters	1.2–4.7 %	structural
14–24 Hz	1.00 of 5 or 6 in five of seven	no; three bands in one campaign	0.3–6 %	local to the sensor position
57–700 Hz	—	—	0.007–0.042 %	mains harmonics

6 Physical interpretation of the identified mode

Sections 4 and 5 are independent of one another: the first uses only the geometry of the foundation, the second only the recorded motion. Combining them tests whether the mechanism the geometry requires is the mechanism the measurements show, and yields two quantities that neither line of evidence could give alone.

6.1 Sway as the governing mechanism

The geometry of Section 4 places rocking 7.3 times above sway on this foundation, and the admitted band is the only low-frequency band in the records. The band is therefore sway. That conclusion is reached without any measurement, and it predicts what the measurements should show.

Sway of a rigid cap on inclined piles has a specific signature, and it is not a pure translation. The cap translates horizontally under the lateral stiffness of the group; but because the axial path through the piles is far stiffer than the lateral one, the same motion also produces a small rotation about the cap centre. The two components are seen by different sensors. A horizontally oriented sensor sees the translation, which is common to every point of a rigid cap, so all horizontal channels move in phase with comparable amplitude. A vertically oriented sensor cannot see the translation at all, and records only the rotation, which appears as opposite edges of the cap moving in antiphase about a nodal line through its centre.

That is what Section 5.2 found, in every setup of both campaigns: horizontal channels in phase, and vertical channels in antiphase about a nodal line. Between corresponding clusters of the two vertical setups the modal assurance criterion is 0.967, over the five points they instrument in common. The prediction and the observation are drawn from disjoint information, one from a setting-out drawing and the other from accelerometer records, and their agreement is the central result of this paper.

Two qualifications apply. The nodal line is located with respect to the instrumented frame, and the position of that frame on the cap is not documented; the identification of its centre with the cap centre rests on the inference set out in Section 3.2. The admitted band, moreover, is not a single mode: its spread within a campaign exceeds the identification scatter by an order of magnitude, and the longest record resolves three distinct poles within it, at 1.073, 1.198 and 1.230 Hz. The simplest explanation is a closely spaced orthogonal pair, resolved unequally by

setups having between three and six channels in one or two directions. Section 5.2 supplies a second indication of the same effect: the two vertical setups of campaign 2 are dominated by different members of the group, so that their spectral peaks disagree even though their corresponding poles have nearly identical shapes. Nothing in the mechanism argument depends on resolving them.

6.2 Equivalent axial pile stiffness

Once the mechanism is fixed, so is the relation between the measured frequency and the axial pile stiffness, and the measurement can be inverted for a quantity the drawings do not contain. Taking the campaign 1 mean of 1.257 Hz gives k_{ax} between 1.7 and 2.7×10^8 N/m per pile, or 0.19 times the elastic stiffness EA/L_{free} of the free-standing length alone.

This value can be checked against an estimate that uses no measured frequency. The shear modulus of each layer follows from its SPT value through the correlation of Ohsaki and Iwasaki [80], $G_0 = 1200\,N^{0.8}$ t/m^2, or about $11.8\,N^{0.8}$ MPa. The correlation was derived from seismic exploration, so it gives the small-strain modulus, which is the modulus that ambient vibration mobilises; no degradation for operational strain is applied. Over the 33.6 m embedded length the modulus averages 86 MPa, against 164 MPa in the layer at the toe. A lower bound of $0.5\,G_0$ is carried as an allowance for the scatter of the correlation and for disturbance of the soft clay by pile installation. The axial stiffness of the embedded length follows from the solution of Randolph and Wroth [4] for a compressible pile in non-homogeneous soil. The pile is long enough for its compressibility to matter, with μL between 3.2 and 4.5, and the rigid-pile limit $2\pi G L_{\mathrm{emb}}/\zeta$ would overstate the stiffness of the embedded length by a factor of 2.4 to 3.3. Placed in series with the free length, the solution gives 4.7 to 5.8×10^8 N/m for an isolated pile.

Pile–soil–pile interaction acts on the embedded length only, since the free length shortens under its own load whatever its neighbours carry. The flexibility of the pile heads is therefore $\mathbf{F} = \mathbf{I}/k_{\mathrm{free}} + \mathbf{A}/k_{\mathrm{emb}}$, with $\mathbf{A}$ the interaction matrix of Section 4.2, $k_{\mathrm{free}} = EA/L_{\mathrm{free}}$ and k_{emb} the stiffness of the embedded length, and the effective stiffness per pile in sway is $\mathbf{v}^{\mathsf{T}}\mathbf{F}^{-1}\mathbf{v}/\mathbf{v}^{\mathsf{T}}\mathbf{v}$. The forward prediction becomes 2.5 to 3.4×10^8 N/m, against 1.7 to 2.7×10^8 N/m from the measurement.

The two ranges overlap, the forward estimate lying between 0.95 and 1.9 times the inverted one. The forward route rests on an SPT-to-modulus correlation and on interaction factors derived for rigid piles, which tend to overstate the interaction between piles as compressible as these. Agreement at that level is as much as the available information supports, and the comparison is an order-of-magnitude check. One narrower conclusion follows: the two estimates overlap only when group interaction is included. Omitting it leaves the forward estimate 1.8 to 3.3 times above the inverted value, even at the lower bound of the soil modulus.

Table 8: Inverted and forward estimates of axial pile stiffness.

Inverted from $f = 1.257$ Hz	1.7–2.7 $\times 10^8$ N/m per pile ($0.19 \times EA/L_{\mathrm{free}}$)
Forward, isolated pile, $G = 0.5$–$1.0\,G_0$ from the SPT profile	4.7–5.8 $\times 10^8$ N/m
Forward, in the group, interaction on the embedded length	2.5–3.4 $\times 10^8$ N/m
Ratio, forward to inverted	0.95–1.9 with interaction; 1.8–3.3 without

6.3 Effect of tower installation

If the mechanism and the stiffness are unchanged between the two campaigns, a change in frequency can come only from a change in participating mass, and the ratio of the frequencies

fixes that change without any model of the soil. This route requires nothing that Section 6.2 had to assume.

The unit of independent observation is the setup, not the individual pole, because several poles originate in the same record. Averaging within setups gives 1.173, 1.240, 1.322 and 1.284 Hz for campaign 1, and 1.133, 1.147 and 1.168 Hz for campaign 2. The two sets are completely separated, the smallest value of the first campaign exceeding the largest of the second, and the mean changes by $-8.41\,\%$. A two-sided permutation test on the setup means returns $p = 2/35 = 0.057$. That is also the smallest value four and three groups can produce, so the separation is as complete as the design allows. Falling short of the conventional level reflects the number of setups, not the size of the effect. Figure 13 and Table 9 give the values. The individual poles overlap between campaigns, and for that reason the analysis is conducted at setup level.

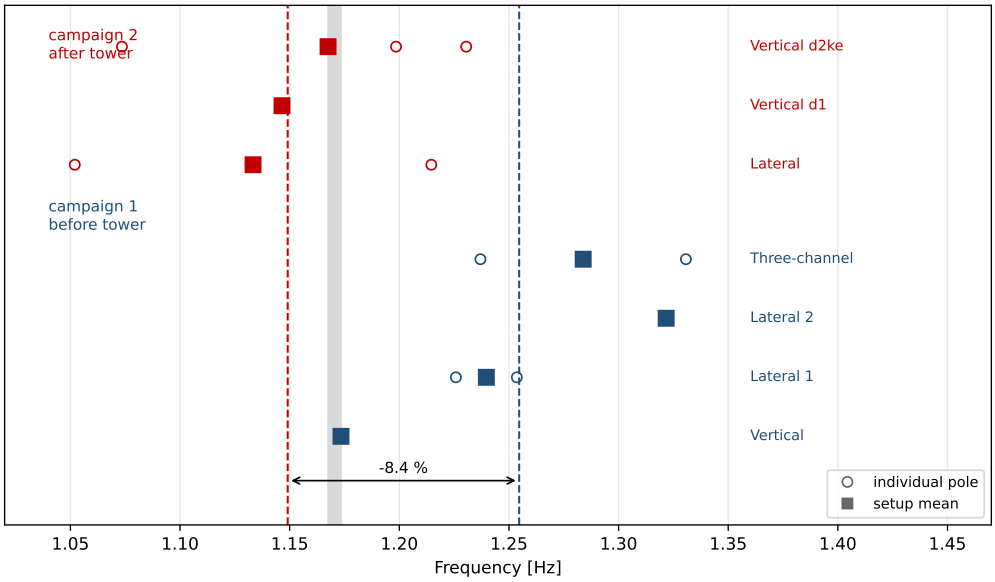


Figure 13: Low-frequency band by setup for both campaigns, showing the individual poles and the setup means. The setup means of the two campaigns do not overlap: the smallest of campaign 1 exceeds the largest of campaign 2, a change in mean of $-8.41\,\%$. The individual poles do overlap, which is why the analysis is conducted at setup level

Table 9: Low-band poles by setup, setup means, and the change between campaigns.

Camp.	Setup	Poles (Hz)	Setup mean (Hz)
1	Vertical	1.173	1.1734
1	Lateral 1	1.226, 1.254	1.2397
1	Lateral 2	1.322	1.3217
1	Three-channel	1.237, 1.331	1.2839
	campaign mean		**1.2547**
2	Lateral	1.052, 1.215	1.1333
2	Vertical d1	1.147	1.1465
2	Vertical d2ke	1.073, 1.198, 1.230	1.1675
	campaign mean		**1.1491**
	change		$-8.41\,\%$

Applying $M_2/M_1 = (f_1/f_2)^2$ to the campaign means implies an additional participating

mass of approximately 950 tonnes, which is the correct order of magnitude for the tower, nacelle and rotor of a turbine of this class.

Three considerations support treating this as the same mode observed twice rather than as two different modes. The deformation pattern measured at the cap is the same before and after installation. A tower fundamental mode would lie near 0.3 Hz by soft-stiff design, a factor of four below the observed band, and the search of Section 5.2 found no acceptable pole anywhere below 0.5 Hz. A change of 8 % is, finally, of the order expected from added mass and not of the order that separates distinct modes. The first of these is measured, the second is an absence rather than a presence, and the third assumes what it supports; the inference is reasonable but not proved. The figure obtained is in any case a participating mass in this mode, not the physical mass of what was installed, since the tower contributes rotational inertia and some stiffness as well as mass.

7 Conclusions

The ordering of sway and rocking in an inclined pile group is a matter of geometry rather than of soil. Because the two mechanisms are resisted by the same axial pile stiffness, that stiffness cancels from the ratio of their frequencies, and what remains depends only on the pile positions, the angle of inclination and the distribution of mass. For any inclination used in construction the separation between the two is wide, so a single low-frequency band identified on such a foundation may be attributed to sway without knowing the ground. The measurements confirm this: the cap translates and rotates in the pattern that sway on inclined piles produces, and the prediction and the observation are drawn from information that does not overlap. The same records also show how readily a measurement of this kind may be misinterpreted. The most prominent poles in them are not modes of the foundation. Some move a single sensor and shift when the sensors are moved. Others decay too slowly to belong to a structure and prove to be interference from the mains supply. Which of them an identification returns depends as much on the settings the analyst chooses as on the foundation itself. Acceptance was therefore made to rest on the properties a mode must possess rather than on how cleanly a pole stabilises. The criterion itself is a first-order result. It indicates which mechanism to expect, not the frequency at which it will appear, and it should be set aside wherever a full impedance model is available. It has been demonstrated on one structure, and showing that it discriminates correctly where the answer is already known would need a documented benchmark. Within those limits it provides a classification obtainable from the setting-out drawing alone.

Statements and Declarations

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Author Contributions Nguyen Hoang Bui: conceptualization, methodology, software, formal analysis, investigation, visualization, writing – original draft. Thi Cam Nhung Nguyen: conceptualization, methodology, investigation, resources, data curation, project administration, supervision, writing – review and editing. Huu-Quyet Nguyen: investigation, data curation. Chung Truong Thanh: software, validation. Hung Nguyen-Dang: writing – review and editing. All authors read and approved the final manuscript.

Data Availability Statement The acceleration records analysed in this study are available from the corresponding author upon reasonable request by e-mail. The design drawings belong

to the project owner and are not redistributed; every quantity read from them is given in Table 1 and Figure 2.

Code Availability The analysis is implemented in Python (NumPy, SciPy, Matplotlib). The code that regenerates every reported quantity, table and figure from the records is publicly available at `https://github.com/chungntu/t18-foundation-oma`.

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